Practice Problems: Calculus

ECON 441: Introduction to Mathematical Economics — Instructor: Div Bhagia

Exercise 6.2

2. Given the function $y = 5x^2 - 4x$:
   (a) Find the difference quotient as a function of x and Δx.
   (b) Find the derivative dy/dx, using the limit definition.
   (c) Find $f'(2)$ and $f'(3)$.
3. Given the function $y = 5x - 2$:
   (a) Find the difference quotient $\Delta y/\Delta x$. What type of function is it?
   (b) Since Δx does not appear in $\Delta y/\Delta x$ in part (a), does it make any difference to the value of $\Delta y/\Delta x$ whether Δx is large or small? Consequently, what is the limit of the difference quotient as Δx approaches zero?

Exercise 7.1

3. Find $f'(1)$ and $f'(2)$ for the following functions:
   (a) $y = f(x) = 18x$
   (b) $y = f(x) = cx^3$
   (c) $f(x) = -5x^{-2}$
   (d) $f(x) = \frac{3}{4}x^{4/3}$
   (e) $f(w) = 6w^{1/3}$
   (f) $f(w) = -3w^{-1/6}$

Exercise 7.2

3. Differentiate the following by using the product rule:
   (d) $(ax - b)\left(cx^2\right)$
   (e) $(2 - 3x)(1 + x)(x + 2)$
7. Find the derivatives of:
   (a) $\left(x^2 + 3\right)/x$
   (b) $(x + 9)/x$
   (c) $6x/(x + 5)$
   (d) $\left(ax^2 + b\right)/(cx + d)$
8. Given the function $f(x) = ax + b$, find the derivatives of:

(a) $f(x)$ (b) $xf(x)$ (c) $1/f(x)$ (d) $f(x)/x$

Exercise 7.3

1. Given $y = u^3 + 2u$, where $u = 5 - x^2$, find dy/dx by the chain rule.
2. Given $w = ay^2$ and $y = bx^2 + cx$, find dw/dx by the chain rule.
3. Use the chain rule to find dy/dx for the following:

   (a) $y = \left(3x^2 - 13\right)^3$ (b) $y = \left(7x^3 - 5\right)^9$ (c) $y = (ax + b)^5$

4. Given $y = (16x + 3)^{-2}$, use the chain rule to find dy/dx. Then rewrite the function as $y = 1/(16x+3)^2$ and find dy/dx by the quotient rule. Are the answers identical?
5. Given $y = 7x + 21$, find its inverse function. Then find dy/dx and dx/dy, and verify the inverse-function rule. Also verify that the graphs of the two functions bear a mirror-image relationship to each other.
6. Are the following functions strictly monotonic? For each strictly monotonic function, find dx/dy by the inverse-function rule.

   (a) $y = -x^6 + 5 \quad (x > 0)$ (b) $y = 4x^5 + x^3 + 3x$

Exercise 10.5

1. Find the derivatives of:

   (a) $y = e^{2t+4}$
   (b) $y = e^{1-9t}$
   (c) $y = e^{t^2+1}$
   (d) $y = 5e^{2-t^2}$
   (e) $y = e^{ax^2+bx+c}$
   (f) $y = xe^x$
   (g) $y = x^2e^{2x}$
   (h) $y = axe^{bx+c}$

3. Find the derivatives of:

   (a) $y = \ln\left(7t^5\right)$
   (b) $y = \ln\left(at^{10}\right)$
   (c) $y = \ln(t + 19)$
   (d) $y = 5\ln(t + 1)^2$
   (e) $y = \ln x - \ln(1 + x)$
   (f) $y = \ln\left[x(1 - x)^8\right]$
   (g) $y = \ln\left(\frac{2x}{1+x}\right)$
   (h) $y = 5x^4 \ln x^2$

7. Find the derivatives of the following by first taking the natural log of both sides:

(a) $y = \dfrac{3x}{(x+2)(x+4)}$ (b) $y = \left(x^2+3\right)e^{x^2+1}$

Exercise 7.4

1. Find $\partial y/\partial x_1$ and $\partial y/\partial x_2$ for each of the following functions:

   (a) $y = 2x_1^3 - 11x_1^2x_2 + 3x_2^2$

   (d) $y = \dfrac{5x_1+3}{x_2-2}$

2. Find f_x and f_y from the following:

   (a) $f(x,y) = x^2 + 5xy - y^3$

   (b) $f(x,y) = \left(x^2-3y\right)(x-2)$

3. From the answers to Prob. 2, find $f_x(1,2)$, the value of the partial derivative f_x when $x = 1$ and $y = 2$, for each function.

5. If the utility function of an individual takes the form

$$U = U(x_1,x_2) = (x_1+2)^2(x_2+3)^3$$

   where U is total utility, and x_1 and x_2 are the quantities of two commodities consumed:

   (a) Find the marginal-utility function of each of the two commodities.

   (b) Find the value of the marginal utility of the first commodity when 3 units of each commodity are consumed.

7. Write the gradients of the following functions:

   (a) $f(x,y,z) = x^2 + y^3 + z^4$

   (b) $f(x,y,z) = xyz$

Exercise 8.1

1. Find the differential dy, given:

   (a) $y = -x\left(x^2+3\right)$

4. Find the point elasticity of demand, given $Q = k/P^n$, where k and n are positive constants.

   (a) Does the elasticity depend on the price in this case?

   (b) In the special case where $n = 1$, what is the shape of the demand curve? What is the point elasticity of demand?

6. Given $Q = 100 - 2P + 0.02Y$, where Q is quantity demanded, P is price, and Y is income, and given $P = 20$ and $Y = 5{,}000$, find the

   (a) Price elasticity of demand.

   (b) Income elasticity of demand.

Exercise 8.2

3. Find the total differential, given

   (a) $y = \dfrac{x_1}{x_1 + x_2}$

4. The supply function of a certain commodity is

$$Q = a + bP^2 + R^{1/2} \quad (a < 0,\ b > 0) \quad [R\text{: rainfall}]$$

   Find the price elasticity of supply ε_{QP}, and the rainfall elasticity of supply ε_{QR}.

5. How do the two partial elasticities in Prob. 4 vary with P and R? In a strictly monotonic fashion (assuming positive P and R)?

6. The foreign demand for our exports X depends on the foreign income Y_f and our price level P: $X = Y_f^{1/2} + P^{-2}$. Find the partial elasticity of foreign demand for our exports with respect to our price level.

7. Find the total differential for each of the following functions:

   (b) $U = 7x^2y^3$

   (f) $U = (x - 3y)^3$

Exercise 8.4

2. Find the total derivative dz/dt, given

(a) $z = x^2 - 8xy - y^3$, where $x = 3t$ and $y = 1 - t$

(b) $z = 7u + vt$, where $u = 2t^2$ and $v = t + 1$

(c) $z = f(x, y, t)$, where $x = a + bt$ and $y = c + kt$

4. Find the partial total derivatives $\S W/\S u$ and $\S W/\S v$ if

(a) $W = ax^2 + bxy + cu$, where $x = \alpha u + \beta v$ and $y = \gamma u$

(b) $W = f(x_1, x_2)$, where $x_1 = 5u^2 + 3v$ and $x_2 = u - 4v^3$

Exercise 8.5

1. For each $F(x, y) = 0$, find dy/dx:

(a) $y - 6x + 7 = 0$

(b) $3y + 12x + 17 = 0$

(c) $x^2 + 6x - 13 - y = 0$

2. For each $F(x, y) = 0$ use the implicit-function rule to find dy/dx:

(d) $F(x, y) = 6x^3 - 3y = 0$

3. For each $F(x, y, z) = 0$ use the implicit-function rule to find $\partial y/\partial x$ and $\partial y/\partial z$:

(a) $F(x, y, z) = x^2y^3 + z^2 + xyz = 0$

Exercise 14.2

1. Find the following:

(a) $\int 16x^{-3}\,dx \quad (x \neq 0)$

(c) $\int \left(x^5 - 3x\right) dx$

(d) $\int 2e^{-2x}\,dx$

Exercise 14.3

1. Evaluate the following:

(a) $\int_1^3 \frac{1}{2}x^2\,dx$

(e) $\int_{-1}^1 \left(ax^2 + bx + c\right)dx$

2. Evaluate the following:

(a) $\int_1^2 e^{-2x}\,dx$

(d) $\int_e^6 \left(\frac{1}{x} + \frac{1}{1+x}\right)dx$

5. Verify that a constant c can be equivalently expressed as a definite integral:

(a) $c = \int_0^b \frac{c}{b}\,dx$

(b) $c = \int_0^c 1\,dt$